

BASRAH INTL, Iraq

WMO: 406890

Lat: 30.549N

Lon: 47.662E

Elev: 11

StdP: 14.69

Time Zone: 3.00 (E03)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	37.4	40.6	19.0	14.3	64.5	22.7	17.1	65.2	32.4	65.4	27.8	63.3	6.0	270	0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.9	120.3	70.7	118.4	70.7	116.5	70.3	80.1	95.7	76.5	95.8	73.8	99.0	13.6	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.8	135.2	88.9	71.5	116.4	86.2	67.6	101.4	82.7	43.8	95.3	40.0	93.9	37.4	97.6	91.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
29.8	25.7	22.2	31.4	123.6	2.6	3.0	29.5	125.7	28.0	127.5	26.6	129.2	24.7	131.4		
			28.6	84.3	2.7	3.2	26.7	86.6	25.1	88.5	23.6	90.2	21.7	92.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	79.5	55.6	60.2	69.5	79.4	91.4	97.6	100.3	99.5	92.4	82.8	66.4	57.3
	DBStd	17.20	5.55	6.27	6.30	6.08	5.39	4.00	3.60	3.40	4.54	5.89	7.46	6.18
	HDD50	29	14	5	0	0	0	0	0	0	0	0	0	10
	HDD65	795	292	156	27	1	0	0	0	0	0	0	71	248
	CDD50	10787	189	290	605	882	1282	1427	1559	1534	1273	1017	493	236
	CDD65	6078	2	21	167	433	817	977	1094	1069	823	553	113	9
	CDH74	99155	11	177	1665	5264	13243	17461	19917	19136	13524	7499	1196	62
	CDH80	70653	0	24	615	2759	9107	13207	15470	14720	9730	4599	419	3
Wind	WSAvg	9.8	8.3	9.3	10.2	10.0	10.3	13.5	12.8	10.5	9.0	8.1	7.8	7.8
Precipitation	PrecAvg	12.4	2.6	1.7	1.9	1.1	0.3	0.0	0.0	0.0	0.0	0.5	1.8	2.5
	PrecMax	17.8	6.5	4.0	5.9	4.0	1.9	0.1	0.1	0.0	0.3	2.2	7.6	6.8
	PrecMin	6.7	0.5	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2
	PrecStd	3.2	1.5	1.0	1.3	0.9	0.4	0.0	0.0	0.0	0.0	0.6	1.8	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.4	82.7	98.5	104.3	115.0	120.6	123.6	122.3	118.3	109.0	93.1	78.9
		MCWB	59.0	59.7	61.7	66.1	69.6	69.9	70.9	70.9	70.6	67.5	64.3	59.0
	2%	DB	71.6	78.9	90.0	100.0	111.4	118.1	120.5	120.2	115.0	104.3	88.2	75.4
		MCWB	57.1	58.3	61.4	65.3	68.4	69.7	70.6	71.2	69.9	66.8	63.5	59.4
	5%	DB	69.4	75.4	86.2	96.4	108.0	115.6	118.5	118.2	112.7	102.1	84.6	72.0
		MCWB	56.7	56.8	60.1	64.2	67.3	69.2	70.6	71.2	69.1	66.6	62.5	58.7
	10%	DB	66.3	72.1	82.5	93.0	105.7	113.2	116.4	116.2	109.7	98.9	80.9	69.5
		MCWB	55.5	56.1	58.8	63.3	66.9	68.5	70.2	70.9	68.4	65.6	62.7	57.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.1	64.8	68.3	70.7	73.7	78.5	81.7	84.9	82.8	79.6	74.2	67.2
		MCDB	69.1	73.4	77.3	86.4	93.2	98.4	96.2	93.6	90.6	88.7	81.5	72.8
	2%	WB	62.6	62.7	66.0	68.9	71.7	72.8	77.3	81.0	76.7	76.5	70.5	64.4
		MCDB	67.1	70.9	77.7	86.5	94.4	106.1	100.5	96.8	96.0	85.4	78.7	70.2
	5%	WB	60.1	60.9	64.1	67.5	70.2	70.9	73.4	76.3	72.8	74.1	67.5	61.6
		MCDB	65.0	69.3	76.7	85.4	96.5	107.3	108.5	103.4	101.3	84.5	76.5	67.6
	10%	WB	57.8	58.8	62.3	66.3	69.1	69.8	71.5	72.9	70.4	71.1	65.3	59.4
		MCDB	63.6	67.8	76.9	84.7	96.6	108.4	111.0	108.8	104.1	85.4	75.7	65.9
Mean Daily Temperature Range	5% DB	MDBR	20.0	21.8	23.7	23.8	25.8	30.9	31.9	33.7	35.0	30.5	24.6	21.4
		MCDBR	22.2	25.0	28.0	27.2	29.0	32.8	34.3	35.4	36.5	33.1	28.1	24.3
		MCWBR	12.1	10.4	9.4	8.7	8.9	9.2	10.1	11.5	11.3	12.8	13.0	12.9
	5% WB	MCDBR	15.7	18.3	21.2	22.1	25.6	30.9	32.7	32.2	33.6	24.8	19.3	16.9
MCWBR		10.6	10.1	9.5	8.5	8.5	9.5	11.5	14.4	13.2	13.3	11.3	11.5	
Clear-Sky Solar Irradiance	taub	0.494	0.532	0.591	0.686	0.679	0.612	0.656	0.605	0.549	0.577	0.531	0.498	
	taud	1.844	1.761	1.632	1.510	1.522	1.608	1.552	1.641	1.742	1.701	1.787	1.847	
	Ebn at Noon	221	227	224	210	213	227	216	225	232	211	207	211	
	Edh at Noon	55	65	79	92	92	84	88	80	69	68	57	52	
All-Sky Solar Radiation	RadAvg	1085	1390	1764	1944	2253	2553	2414	2266	1991	1536	1157	1005	
	RadStd	73	81	69	104	108	70	69	57	66	68	83	91	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	+0.52	N/A	N/A	+0.79	N/A	N/A	N/A	N/A	+153	+149

Nomenclature: See separate page